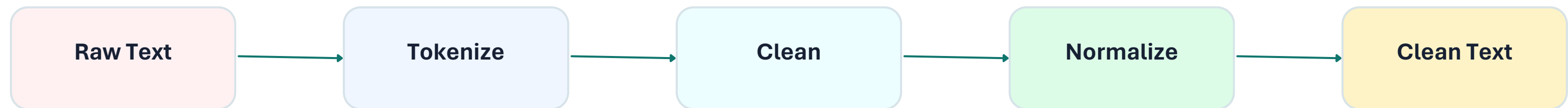


Day 1

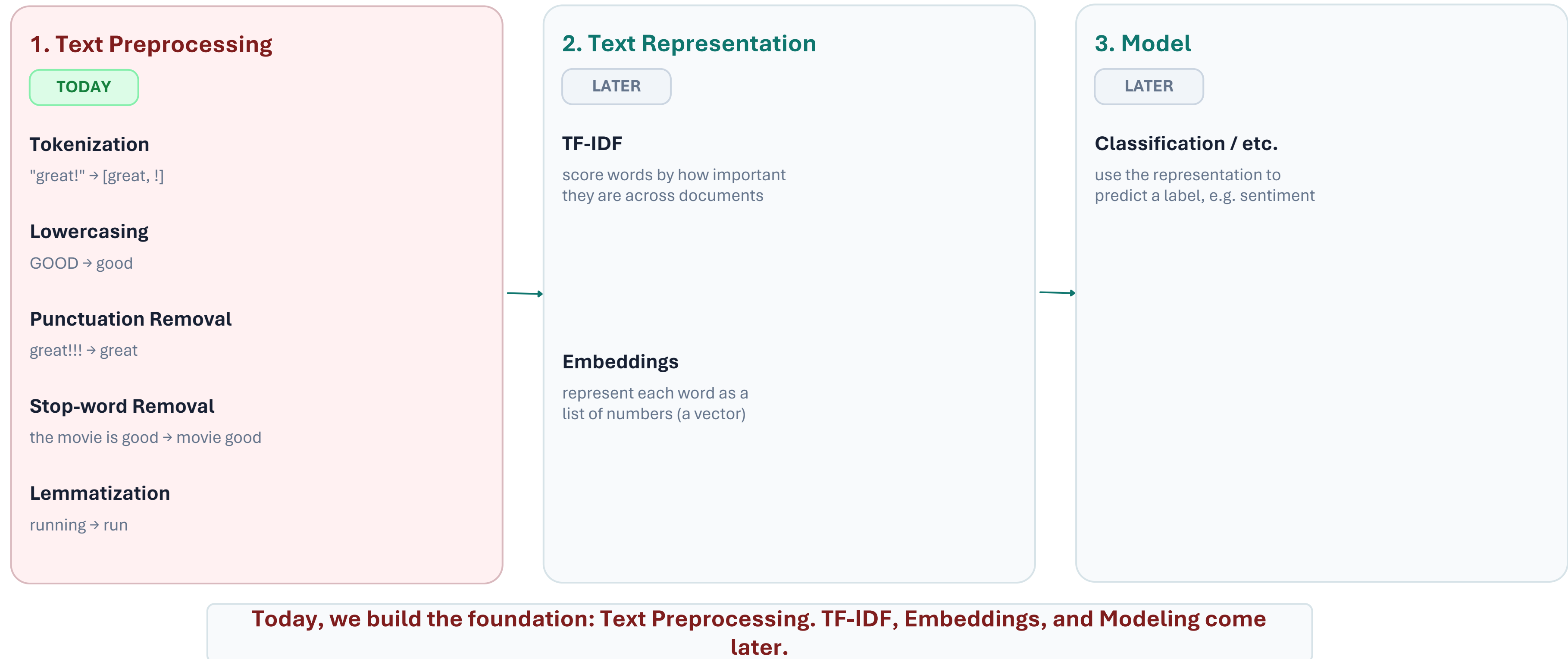
NLP & Text Preprocessing

Core concepts explained visually



The Full NLP Pipeline

Where today's concepts fit in the bigger picture

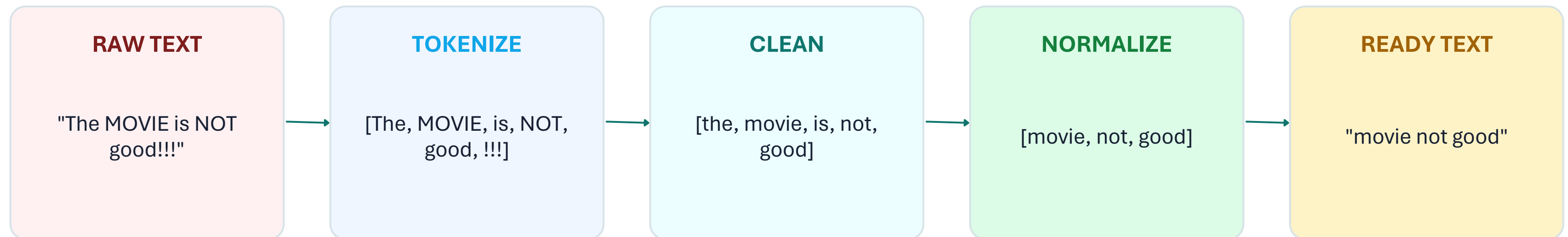


The Big Picture

What Day 1 is really about

One idea to remember

Human text is meaningful to us — but messy and inconsistent for a machine.



Goal: remove unnecessary variation — without removing the meaning the task

1. NLP & Raw Text

Start by understanding what enters the pipeline

Human view

“This movie was absolutely amazing!”

We immediately understand the meaning.

Meaning

Machine view

CAPS MOVIE

Punctuation !!!

Numbers 10/10

HTML

Contractions wasn't

Raw text = text exactly as it arrives, before preprocessing.

2. Tokenization

Split text into smaller processable units

What is a token?

A token is a unit of text. It can be a word, a sub-word, or sometimes a character.

Word tokenization

The movie was great!

The

movie

was

great

!

Each full word becomes a token.

Sub-word tokenization

playing



play

ing

Useful when a full word is unknown but its smaller pieces are known.

Tokenization SPLITS text — it does not clean it.

3. Cleaning & Normalization

Reduce unnecessary variation in the text



Cleaning is a decision: remove noise, preserve useful information.

4. Stop-word Removal

Common words can be low-signal — but not always

Typical stop words

the

is

a

this

was

They may add little signal in some tasks.

Goal: reduce low-value words.

The negation trap

this movie is not good



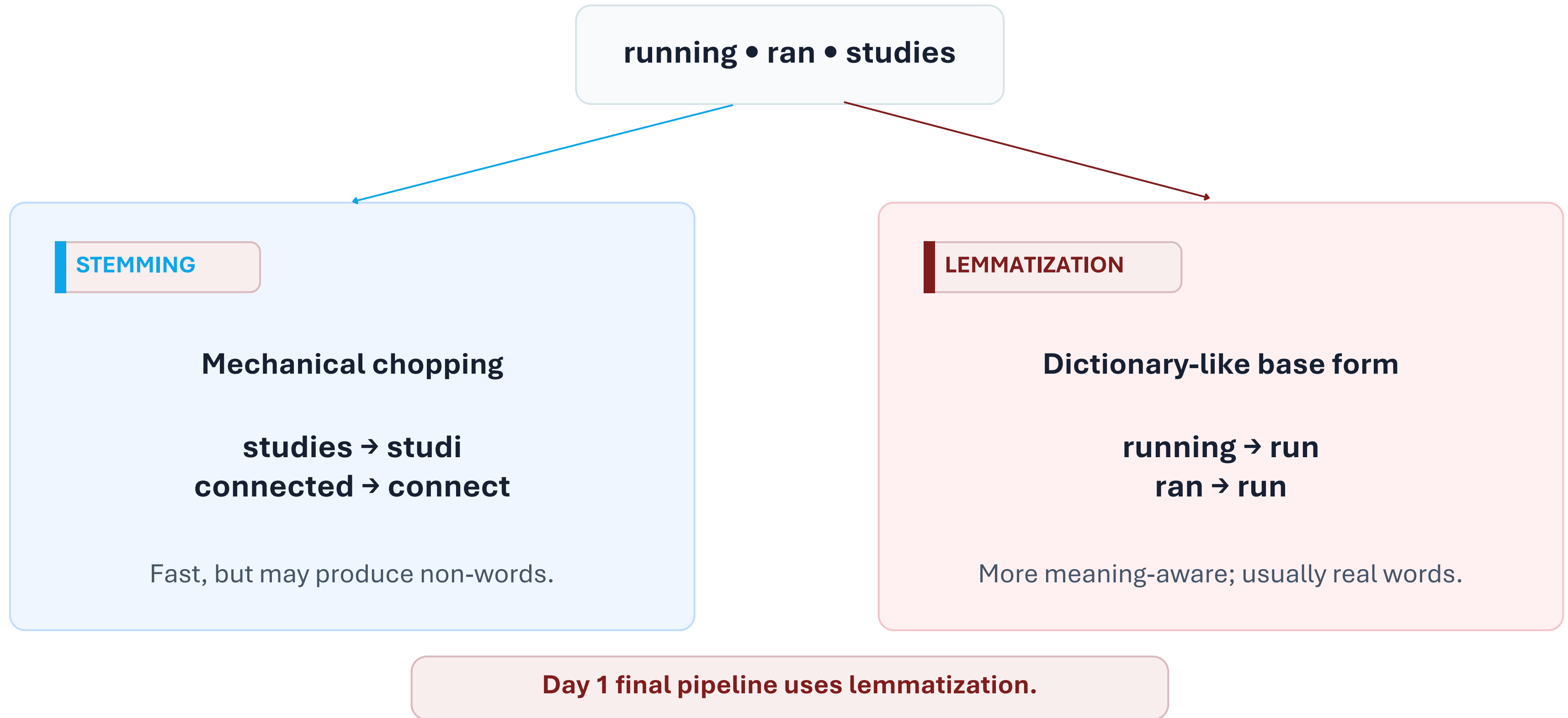
movie good

Meaning changed!

Task-aware rule: keep “not” when sentiment depends on it.

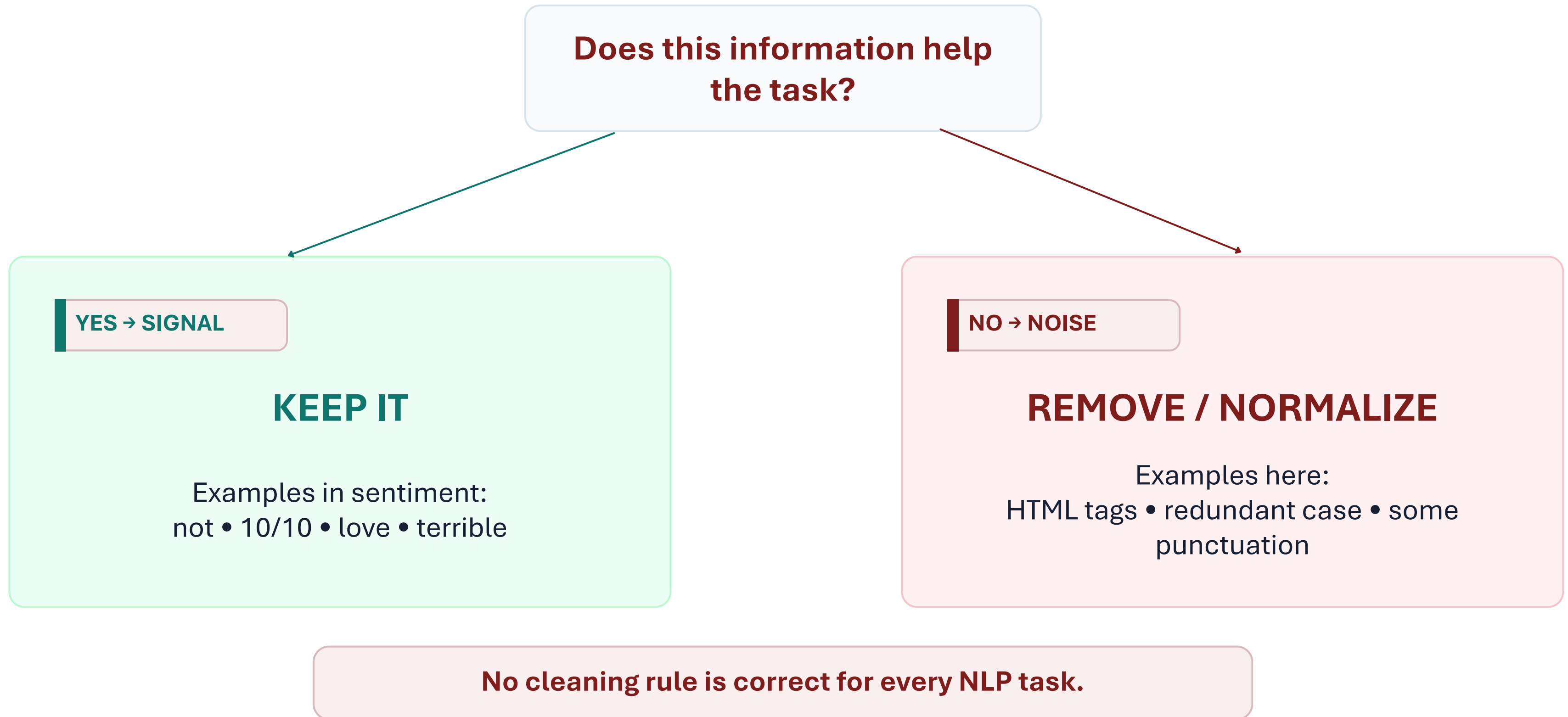
5. Lemmatization vs. Stemming

Both reduce word variation — but in different ways



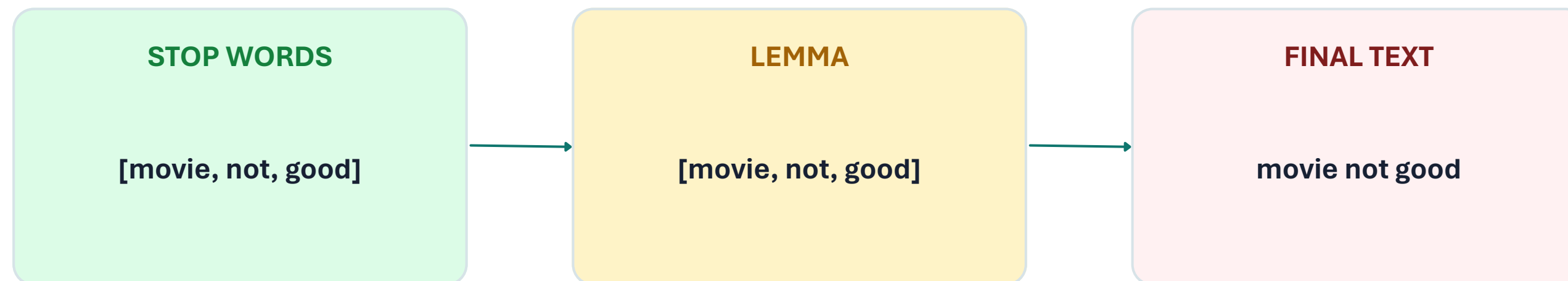
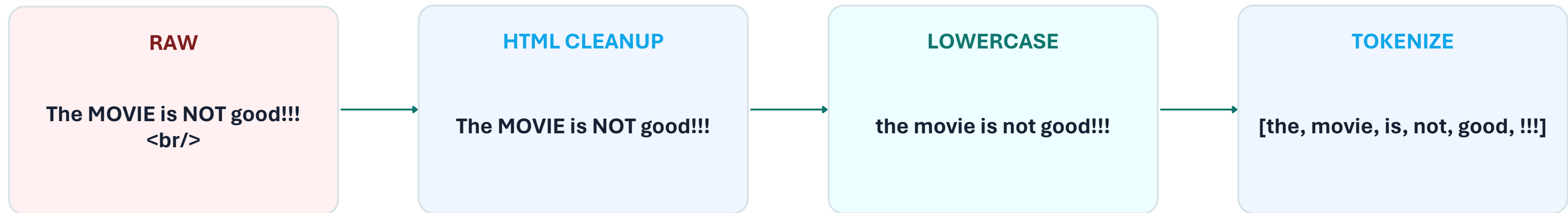
6. Noise vs. Signal

The most important decision in preprocessing



7. The Complete Day 1 Pipeline

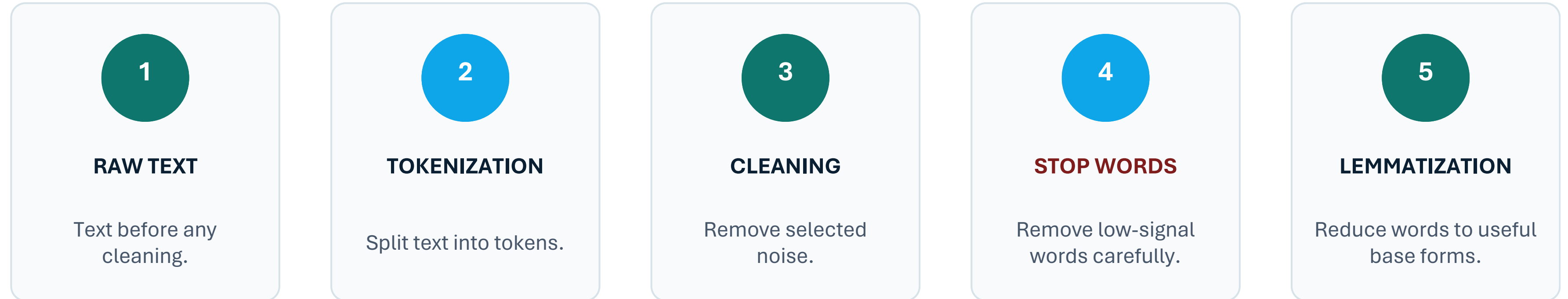
See every concept working together



Every step has a different job — but the same goal: useful, consistent text.

Day 1 Mental Map

A one-slide revision of the core concepts



THE GOLDEN RULE

Preprocessing is task-dependent: remove noise, preserve signal.

If a cleaning step changes the meaning your task needs, it is the wrong cleaning step.